S03. Effective conductivity bounds for two-phase composites

Pure Thought research portfolio

Revised 17 September 2026

S03. Effective conductivity bounds for two-phase composites

Source: synthesis challenge 13; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

Mix two materials that conduct heat or electricity differently. The overall conductivity depends on how the materials are arranged, not just how much of each is present. This problem asks for reliable upper and lower bounds on the conductivity of a specified repeating arrangement, and a careful distinction between its performance and limits that apply to every arrangement.

Key terms

- **Conductivity** — A coefficient relating a driving gradient to the resulting current or heat flux in the selected linear model.
- **Two-phase composite** — A material made from two constituent material types.
- **Volume fraction** — The fraction of space occupied by one constituent.
- **Effective conductivity** — The large-scale response obtained after averaging the microscopic structure.
- **Conductivity tensor** — A matrix allowing that response to depend on direction.
- **Potential** — A scalar field, such as electric potential, whose gradient drives transport.
- **Flux** — The local flow of the transported quantity per unit area.
- **Divergence-free flux** — A steady flow with no internal accumulation or sources.
- **Variational bound** — An inequality obtained by optimizing or testing an energy principle.

Why this matters

Composite design seeks useful bulk properties from available ingredients. Bounds reveal whether a proposed arrangement performs well, whether a computation is accurate, and how much improvement remains possible. Simple layered structures also provide exact reference cases for checking more complicated methods.

What progress would mean

Progress would produce a certified interval for a chosen directional response and clarify which bounds are geometry-specific or universal. The result would concern the stated positive, linear conductivities; nonlinear effects, interfaces and manufacturing constraints would need separate treatment.

Revised core question

For fixed positive phase conductivities and volume fraction, can rigorous effective-conductivity bounds and matching constructions be obtained in a specified microstructure class?

Precise scope, target and outputs

Choose scalar conductivities a_1,a_2>0, periodic unit cell Y, phase fraction f, and a unit direction e. Define e^TA_eff e=min_{periodic u}⟨a|e+∇u|^2⟩. Use an admissible potential trial field for an upper bound and a divergence-free flux variational principle for a lower bound. Distinguish bounds valid for every arrangement from optimization over a finite geometry family. Fix isotropy assumptions before comparing isotropic bounds. Cross-property elasticity/thermal results are separate extensions with both constitutive laws specified.

Detailed critique of the synthesis framing

The original "universal bounds" problem spans conductivity, permittivity and elasticity, including tensor and frequency effects with different variational principles. "Beyond Hashin–Shtrikman" is not a measurable novelty requirement until the symmetry class and available information are fixed. A finite set of candidate microstructures cannot characterize the unrestricted G-closure. The revision makes one scalar static problem precise and makes the bound direction and construction obligation explicit.

Validation and milestones

Baseline: verify arithmetic and harmonic directional bounds with layered composites. Strong: matching bounds for one declared geometry class or a certified gap to the unrestricted bound. Check volume fraction and ellipticity exactly; report discretization and geometry errors separately.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$e \cdot A^* e = \min_{\phi \text{ periodic}} \langle a(x)|e + \nabla\phi|^2\rangle, \qquad a(x) \in \{a_1, a_2\},$$

$$\left(\frac{f}{a_1} + \frac{1-f}{a_2}\right)^{-1} |e|^2 \le e \cdot A^* e \le (f a_1 + (1-f)a_2)|e|^2.$$

Assume positive scalar phases, fixed volume fraction f, and a normalized cell average. Potential trials give upper bounds and the corresponding flux principle gives lower bounds. A finite microstructure list does not characterize an unrestricted G-closure.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.